A State-Aware Zero Trust Communication Approach for Offshore Wind Farm*

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Abstract

Because physical on-site access is constrained, offshore wind farm OT environments rely heavily on remote monitoring, maintenance, and control, and consequently diverse communications with different operational purposes and levels of importance continuously occur. However, existing Zero Trust discussions in OT environments have mainly focused on structural aspects such as access restriction, path control, and segmentation, and have not sufficiently specified how the operational significance of communication should be interpreted or how it should be connected to differentiated protection in OT environments. Accordingly, this paper proposes a state-aware communication approach that classifies offshore wind farm OT communication into normal, warning, and critical states, interprets the meaning and importance of communication differently according to the operational state, and then applies differentiated protection levels accordingly. The controlled prototype-level case study results show that the proposed approach maintained the loss rate of important communication at 0.3% while limiting the total communication volume to 4.1 MB, thereby forming a more balanced protection structure between the visibility of important communication and the overall communication burden.

1 Introduction

Offshore wind farm OT(Operational Technology) environments, as illustrated in Fig. 1, rely heavily on remote monitoring, maintenance, and control because physical on-site access is constrained, and consequently diverse types of communication continuously occur[1]. Operational data generated at lower layers are repeatedly delivered to upper operational layers, and maintenance and control are likewise performed through remote paths[2]. NIST SP 800-82 Rev.3[3] emphasizes that OT security

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must consider performance, reliability, and safety requirements that differ from those of general IT, while the U.S. Department of Energy (DOE) states that distributed energy resource (DER) systems should be designed, built, and operated according to a zero-trust model. Therefore, communication in offshore wind farm OT environments should be examined not merely as a connectivity issue but from the perspective of zero trust communication[3][4]. However, offshore wind farm OT communication does not have a single, uniform character. Some communications correspond to repetitive operational telemetry transmission, while others are directly related to maintenance, management, or actual control. In addition, communication in wind power plants may be implemented not only through wired networks but also through wireless and cellular means; thus, not only functional differences but also instability in communication conditions must be taken into account.

The core of zero trust does not lie in uniformly applying the strongest protection to all communications, but in assigning the minimum necessary level of protection through policy after individually evaluating each communication without implicitly trusting any of them[5]. However, existing discussions have primarily focused on structural aspects such as access restriction, path control, and segmentation, and have not sufficiently specified how the operational significance of communication in offshore wind farm OT environments should be interpreted or how it should be connected to protection differentiation. If the same lightweight protection is applied to all communications, protection for important communications may be insufficient; conversely, if the same strong protection is applied to all communications, unnecessary overhead may accumulate. Therefore, zero trust communication in offshore wind farm OT environments needs to be concretized not as uniform protection, but as an approach that interprets the significance of communication differently according to the operational state and applies different protection levels accordingly.

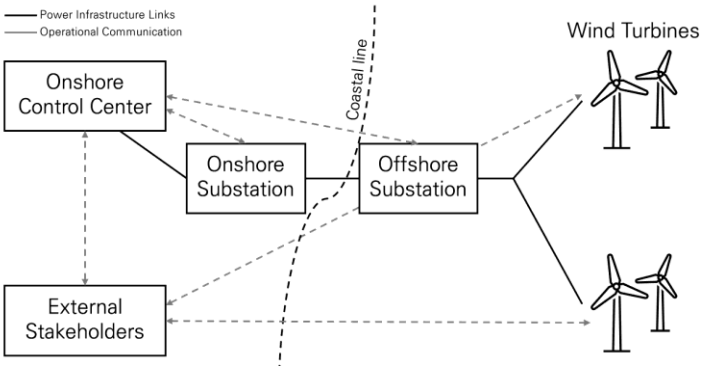


Figure 1: Conceptual Representation of an Offshore Wind Farm

The research questions of this paper are as follows:

- In offshore wind farm OT environments, according to what criteria should zero trust communication distinguish the significance of each communication and the required protection level?
- In offshore wind farm OT environments, is it possible to interpret the significance of communication differently according to the operational state and apply different protection levels accordingly?

The novelty of this paper does not lie in proposing context-aware protection itself. Rather, it lies in applying operational-state-dependent communication criticality to offshore wind farm OT communication. Existing dynamic Zero Trust studies mainly adjust access policies according to user,

device, network, or segment context, whereas this study focuses on how the operational state of a physical asset changes the meaning of the same communication and the required protection level. In this sense, the proposed approach complements existing Zero Trust architectures by linking operational state, communication criticality, and differentiated protection in a unified communication-handling model.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents a state-aware communication approach for zero trust communication in offshore wind farm OT environments. Section 4 explains the applicability of the proposed approach through a prototype-level case study using a representative operational variable. Finally, Section 5 concludes the paper and discusses future research directions.

2 Related Works

Existing studies have proposed various structures for applying Zero Trust to OT environments by pointing out the limitations of the traditional perimeter-based security model and focusing on continuous authentication and authorization, least privilege, policy-based access control, micro-segmentation, remote access control, and context-aware dynamic policies.

There are studies that attempt to reconstruct OT architectures themselves from a Zero Trust perspective. These studies sought to design Zero Trust structures that can be practically applied to OT environments through elements such as Purdue model enhancement, asset identification, policy-based access control, legacy system protection, and continuous monitoring. In particular, studies targeting smart manufacturing and legacy manufacturing environments reflected practical constraints such as end-of-support operating systems, long-life operational equipment, and the difficulty of patch application, and proposed not merely theoretical Zero Trust models but operationally deployable security structures [7], [10], [11].

There are also studies that attempt to implement Zero Trust in industrial networks and IIoT infrastructures with a focus on micro-segmentation and remote access protection. Studies such as [6], [13], and [15] sought to segment industrial networks more finely and control resource access more precisely through SDN, per-device segmentation, policy enforcement points, and cross-layer access control. In addition, [9] and [12] proposed architectures that verify the access of external entities on a per-session basis and grant segmented access privileges in industrial remote access environments.

More recently, studies have also emerged that seek to extend Zero Trust toward context-aware and dynamic Zero Trust. [14] proposed a structure that adjusts access policies in real time by reflecting user behavior, device status, network conditions, and segment criticality, while [15] emphasized multi-layer trust evaluation that considers interactions between the cyber layer and the physical layer in industrial CPS. [8] presented a stepwise structure of security zone formation, device authentication, and cryptographic negotiation in wireless IIoT environments, thereby translating Zero Trust requirements into a more concrete communication-technology level.

Although prior studies provide valuable mechanisms for identity verification, segmentation, policy enforcement, remote access control, and context-aware policy adjustment, their main focus remains on deciding whether an entity should be allowed to access a resource under a given context. They do not sufficiently address how the operational state of a physical asset changes the semantic importance of telemetry or control-related communication itself. This distinction is important in offshore wind farm OT environments, where the same communication may function as routine reporting during normal operation, as evidence for abnormal-state interpretation during warning conditions, or as a basis for control-oriented response under critical conditions. Therefore, this study complements prior dynamic Zero Trust approaches by focusing on operational-state-dependent communication criticality rather than only access permission or segmentation.

3 State-Aware Zero Trust Communication Approach in Offshore Wind Farm OT Environments

This section presents a state-aware communication approach for concretizing zero trust communication in offshore wind farm OT environments. Because it is difficult to treat all communications in offshore wind farm OT environments as having the same significance and requiring the same protection level, this paper proposes an approach that interprets the significance of communication differently according to the operational state and applies different protection levels accordingly.

In the proposed structure, the operational state is used as a contextual input for Zero Trust policy decisions. The state evaluation logic continuously interprets operational data and provides the current state to the policy decision function, which considers entity identity, communication purpose, and the required protection level. The corresponding policy enforcement function then applies state-specific communication handling, such as lightweight reporting, sequence-number-based reinforced reporting, or a restricted protected control session. In this way, the proposed approach complements existing OT Zero Trust architectures by adding operational-state awareness to the policy decision and enforcement process.

This study assumes that communication paths between offshore OT components and upper operational functions are not implicitly trusted. An adversary may attempt packet-loss or denial-of-service-like disruption, replay, spoofed telemetry, message injection, or man-in-the-middle interception. The proposed approach is intended to address these risks at the communication-policy level by combining state-dependent protection selection, entity identity verification, policy-based enforcement, sequence tracking, and restricted protected sessions. However, compromise of local turbine safety controllers, root trust anchors, or physical destruction of field assets is outside the scope of this prototype-level study, and active adversarial validation remains future work.

3.1 Limitation of Uniform Protection

In offshore wind farm OT environments, repetitive operational data reporting, maintenance communication, management signaling, and control-related communication coexist. Since these communications differ in their purpose and operational consequences, it is not appropriate to apply the same protection procedure uniformly to all of them. For example, if all communications are handled only with lightweight protection, protection may be insufficient for operationally more important abnormal-state intervals or control-related communication. Conversely, if the same strong protection is applied to all communications, unnecessary communication burden may accumulate in the repetitive reporting structure.

Therefore, zero trust communication in offshore wind farm OT environments should be understood not simply as a matter of applying stronger protection more broadly, but rather as a matter of first determining under which state a given communication should be interpreted as more important and then selecting an appropriate protection level accordingly. Reflecting this perspective, this paper uses the operational state as the basis for communication protection differentiation.

3.2 State-Aware Communication Operation

This paper classifies offshore wind farm OT communication into three operational states: normal, warning, and critical. Here, state does not refer to the technical property of the communication itself, but rather to a concept that reflects the operational significance of the communication and the required protection level under the current operating condition. In other words, even the same type of

communication may serve as routine reporting during normal operation, require higher visibility and reliability when abnormal signs appear, and demand stronger protection for actual control intervention under critical conditions.

Table 1 summarizes the meaning of each state used in this paper and the corresponding direction of basic communication handling. The purpose of this table is not to define absolute thresholds based on a specific industrial standard, but to provide a conceptual basis for explaining state-aware communication operation in offshore wind farm OT environments. Specific numerical thresholds are illustratively defined in the case study using a representative operational variable.

The normal state refers to a condition in which the equipment is operating within the normal range

State	Meaning	Communication Characteristics	level of protection
Normal	Normal Operation	Periodic Reporting	Lightweight protection
Warning	Warning Signals	Reinforced Reporting	Enhanced Protection
Critical	Critical Condition	Control Intervention	Strong Protection

Table 1: Definition of Operational States

and communication mainly serves the role of routine telemetry reporting. Since continuous upward reporting is required in this interval, this paper adopts UDP-based lightweight reporting as the default in order to reduce communication burden. In other words, in the normal state, the focus is not on uniformly applying high-cost protection procedures to all reporting traffic, but on transmitting repeatedly generated operational data with low overhead.

The warning state refers to a condition in which abnormal signs are observed and greater attention is required for subsequent state assessment and operational response. As explained in the introduction, this interval requires consideration of environments in which packet loss may occur, such as wireless communication. Accordingly, this paper applies a method that assigns sequence numbers to reported data in order to ensure continuity in monitoring and tracking. In other words, the warning state goes beyond maintaining simple routine reporting and serves as a phase that reinforces the visibility and traceability of communication required for interpreting abnormal conditions.

The critical state refers to a condition in which actual operational intervention or control-oriented response is required. In this interval, the impact of communication failure or spoofing becomes greater, and therefore this paper applies the highest level of protection. In other words, the critical state is interpreted not as a monitoring-oriented communication condition, but as one that requires control-oriented communication through a restricted protected channel.

Fig. 2 shows the proposed flowchart of the state-aware communication operation. The overall flow consists of operational data collection, state evaluation, state-specific communication handling, and repeated state updates. First, operational data generated at lower layers are delivered to upper operational functions, and the system determines the current operational state based on these data. In the prototype case study, this state transition is implemented as a rule-based evaluation that compares the generator bearing temperature with the illustrative thresholds defined in Section 4.1. Different protection levels are then applied according to the identified state. Since the state-evaluation layer directly affects protection-level selection, deployable implementations should not rely on unauthenticated telemetry alone. Telemetry used for state evaluation should be bound to authenticated device identities and protected by integrity checks, sequence tracking, and audit logs; cross-checking multiple operational variables or independent monitoring sources can further reduce the risk of state-suppression attacks, such as spoofed normal telemetry during an intrusion.

In the normal state, routine telemetry reporting is the main form of communication, and therefore this paper applies a lightweight protection method that maintains low communication burden. The purpose of this interval is not to process all reports through high-cost protection procedures, but to sustain a continuous data flow during normal operation.

In the warning state, since the possibility of abnormal conditions is recognized, a higher level of communication assurance is required than in the normal state. Accordingly, sequence numbers are applied in this interval so that important reported data can be retained more reliably. This does not mean that all communication is immediately converted into a strong control channel, but rather that the information serving as the basis for state assessment and subsequent response is secured more reliably.

In the critical state, since the possibility of actual control intervention becomes higher, the significance of communication is also interpreted as the highest. Therefore, this paper assumes that communication in the critical state is carried out through a restricted protected channel to which strong protection is applied. In the proposed operation, the critical-state protected session is established only after the system identifies a critical condition and the policy logic approves the need for control-oriented communication. The session is limited to the approved communication purpose and is revoked after the required control-related task is completed. Therefore, the critical state is not treated as a permanent high-protection mode, but as a task-scoped and temporary communication condition for protected control communication.

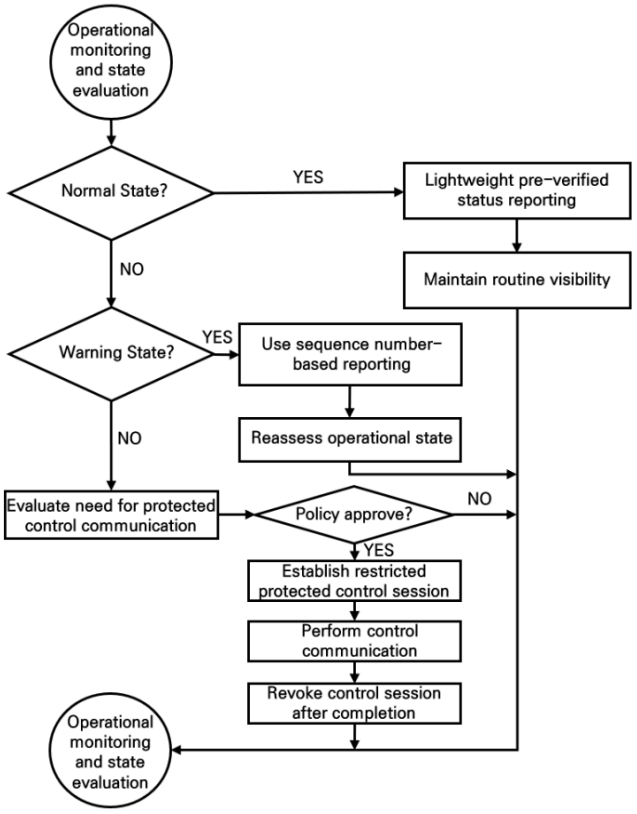


Figure 2: Flowchart of the State-Aware Communication Operation

In this way, the state-aware communication approach proposed in this paper does not apply the same protection to all communication. Instead, it interprets communication significance differently according to the state and adjusts the protection level accordingly. Such a structure is consistent with the basic zero trust principle of assigning necessary protection according to context without relying on implicit trust.

4 Case Study

This section uses one representative operational variable of the offshore wind farm OT environment to illustratively define the normal, warning, and critical states, and demonstrates, through a prototype-level case study, how the state-aware communication approach can be applied in practice. The purpose of this case study is not to precisely reproduce the entirety of a real offshore wind farm, but rather to illustratively examine whether an approach that interprets the significance of communication differently according to the operational state and applies different protection levels accordingly can be implemented.

4.1 Case Study Configuration

In this prototype-level case study, generator bearing temperature was selected as a representative operational variable of an offshore wind farm OT environment. This variable is commonly associated with equipment condition monitoring and is suitable for illustrating how operational states can be mapped to different communication-handling policies. The purpose of using this variable is not to define a complete turbine health assessment model, but to construct a controlled scenario in which normal, warning, and critical states can be distinguished in an intuitive manner.

Specifically, the bearing temperature was defined as normal when it was 60°C or below, warning when it was between 61°C and 80°C, and critical when it was above 80°C. These values are not intended to represent universal turbine safety limits, vendor-specific alarm thresholds, or industrial standards. They are used only as illustrative thresholds for explaining state-aware communication operation. Accordingly, the 60°C and 80°C values should not be interpreted as literature-derived turbine condition-monitoring thresholds; in actual deployments, such thresholds should be determined from turbine-specific historical data, vendor alarm settings, or condition-monitoring guidelines. The reporting interval was set to one second for all schemes.

The experimental environment was configured as a controlled communication-stress scenario that reflects simplified wireless communication instability in offshore wind farm environments. As offshore wind farms move farther from the coast, wireless or cellular communication links may become more important for remote monitoring and operation. To reflect this condition in a simplified prototype setting, the packet-loss condition was modeled as independent random loss with a fixed probability of 5% for each reporting packet. The same topology, reporting schedule, observation window, telemetry message format, and packet-loss condition were applied to all schemes so that the differences in warning-state loss and total communication volume could be attributed to the protection strategy rather than to different traffic-generation conditions. In terms of loss handling, r0 did not apply any additional recovery mechanism, r1 applied non-selective retransmission without considering the operational state, r2 applied sequence-number-based recovery to all reporting traffic, and the proposed scheme applied sequence-number-based reinforced handling selectively to warning-state reports while using a restricted protected session for critical-state communication. Therefore, the case study should be interpreted as a controlled prototype-level comparison, not as a general validation of all offshore wind farm communication environments.

This setting represents non-malicious operational network stress caused by simplified wireless communication instability, rather than a full active-adversary experiment. Active attack profiles such as denial-of-service, message injection, or man-in-the-middle interception require separate adversarial validation and are therefore treated as future work in this prototype-level study.

4.2 Comparison Targets

As shown in Table 2, this case study compares four communication schemes, each representing a distinct protection strategy. The purpose of this comparison is to examine the trade-off between

Scheme	Routine Reporting	Loss Handling	Critical State Response
r0	Lightweight reporting	None	No separate procedure
r1	Lightweight reporting	Non-selective retransmission	No separate procedure
r2	Sequence-numbered reporting	Missing-report recovery	No separate procedure
Proposed	Lightweight reporting	Warning-state retransmission	Restricted control session

Table 2: Comparison of Communication Schemes

communication burden and warning-state visibility under the same controlled experimental condition. These schemes are defined as analytical baselines for comparing protection strategies rather than as vendor-specific implementations or standardized industrial profiles.

r0 represents a lightweight-only baseline. It performs UDP-based lightweight reporting in all states without additional loss handling or state-dependent reinforcement. This scheme is used to represent the lowest communication-burden case, but it does not provide additional visibility or recovery when abnormal signs appear.

r1 represents a non-selective reinforcement baseline. It maintains lightweight reporting as the default communication method, but applies retransmission-based reinforcement without considering the operational state. This scheme is used to examine whether reinforcement that is not guided by state awareness can effectively reduce warning-state data loss.

r2 represents a uniform reinforced recovery baseline. It applies sequence-number-based recovery to all reporting traffic regardless of whether the current state is normal, warning, or critical. This scheme is used to represent a strong but non-selective protection strategy that improves data retention at the cost of increased communication burden.

The proposed scheme represents the state-aware hybrid protection strategy described in Section 3. It applies UDP-based lightweight reporting in the normal state, sequence-number-based reinforced reporting in the warning state, and restricted protected communication in the critical state. Unlike r2, it does not reinforce all reporting traffic uniformly. Instead, it selectively applies additional protection when the operational state indicates that communication visibility or control-oriented protection is required.

4.3 Evaluation Metrics

This case study used two metrics. The first metric is the warning-state prioritized data delivery loss rate, which indicates the proportion of warning-state report data that failed to be delivered within the intended reporting window. This metric is used to evaluate how effectively each scheme preserves important communication during warning-state intervals. Fig. 3 shows the warning-state prioritized data delivery loss rate for the four schemes.

The second metric is the total communication volume over 10 minutes. This metric indicates the communication burden introduced by each scheme under the same experimental condition. It is used to compare whether each protection strategy increases communication overhead while attempting to improve warning-state visibility. Fig. 4 shows the total communication volume over 10 minutes for the four schemes.

4.4 Case Study Results

As shown in Fig. 3, the warning-state prioritized data delivery loss rate was 4.5% for r0, 5.1% for r1, 0.3% for r2, and 0.3% for proposed. In other words, r2 and proposed showed the lowest loss rates in the warning state, whereas r0 and r1 showed relatively higher loss rates. This indicates that simply relying on lightweight reporting alone may not be sufficient to ensure the retention of important

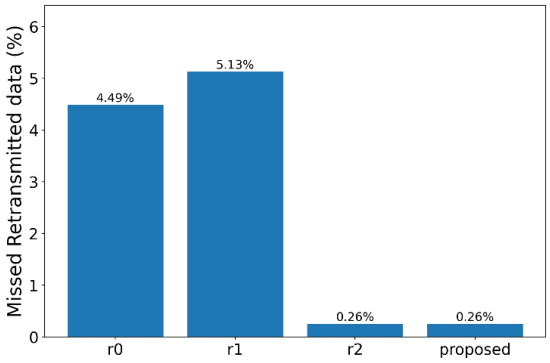


Figure 3: Warning-state prioritized data delivery loss rate under the 5% packet-loss condition

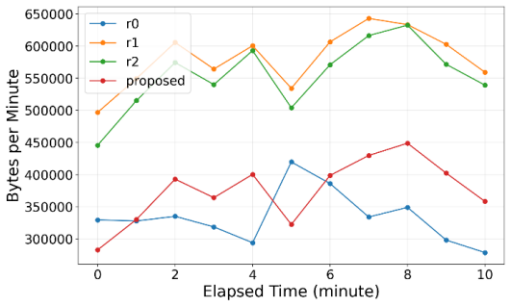


Figure 4: Total communication volume over 10 minutes under the 5% packet-loss condition

communication in abnormal-state intervals, and that non-selective reinforcement without state awareness may also be ineffective. In particular, proposed achieved the same level of warning-state visibility as r2 by applying sequence-number-based reinforced handling only when necessary in the warning state.

As shown in Fig. 4, the total communication volume over 10 minutes was 3.7 MB for r0, 6.4 MB for r1, 6.1 MB for r2, and 4.1 MB for proposed. Although r0 showed the lowest communication burden, its warning-state visibility was insufficient. Because r1 and r2 applied broader recovery handling, their communication usage increased significantly. In contrast, proposed selectively applied only the reinforcement needed in the warning state, thereby achieving the same level of warning-state reliability as r2 while maintaining a lower overall communication volume.

Taken together, r0 had the lowest burden but lacked state-aware handling, r1 incurred increased overhead due to non-selective reinforcement without state distinction while failing to sufficiently reduce warning-state loss, and r2 showed a low warning-state loss through the strongest level of uniform reinforcement but increased the communication burden by reinforcing even routine reporting. By contrast, proposed can be interpreted as a state-aware hybrid structure that lies between these schemes, maintaining low burden in the normal state while reinforcing the visibility and retention of important communication in the warning state and considering a stronger protected channel in the critical state.

4.5 Discussion

The main finding of this case study is that state-aware hybrid protection can form a more flexible protection structure than a single uniform protection scheme in offshore wind farm OT communication. Although r0 showed the lowest communication overhead, its reliability in the warning state was insufficient. Although r1 applied non-selective reinforcement, it still failed to adequately reduce warning-state loss. r2 achieved a low warning-state loss rate, but its communication burden increased because the same reinforced procedure was applied to all reporting traffic. By contrast, the proposed scheme maintained the warning-state loss rate at the same level as r2, 0.3%, while limiting the total communication volume to 4.1 MB, which was lower than r2's 6.1 MB. This indicates that the proposed

scheme can provide a more balanced trade-off between routine reporting efficiency and warning-state communication visibility.

These results are directly connected to the problem statement raised in the introduction. Offshore wind farm OT communication does not have a single, uniform character. Routine telemetry, abnormal-state reporting, and control-oriented communication have different operational meanings and different security requirements. Therefore, it is not appropriate to uniformly trust or assign the same protection level to all communication based only on network location or boundary membership. The proposed scheme addresses this issue by interpreting the significance of communication according to the operational state and applying differentiated protection levels accordingly. In particular, it maintains lightweight reporting in the normal state, reinforces visibility and traceability in the warning state, and applies a restricted protected session in the critical state.

The critical-state handling should be interpreted as an important part of the proposed state-aware communication structure. In the prototype implementation, the critical state was not treated merely as a conceptual label, but as a condition in which a restricted protected communication session is established for control-oriented communication. The session is designed to be limited to an approved communication purpose and to be revoked after the required task is completed. This demonstrates that the proposed state-aware approach can be extended beyond monitoring-oriented reporting and can support a stronger communication mode when operational intervention is required. However, this result should still be understood as a prototype-level validation of the restricted protected session mechanism, not as a complete validation of all industrial real-time control requirements. In actual offshore wind farm OT environments, additional validation would be required to measure session establishment delay, control-message latency, revocation timing, and compatibility with site-specific safety requirements.

The restricted protected session in the critical state is not intended to replace local safety interlocks or hard real-time emergency shutdown mechanisms. In OT environments, emergency protection functions that require deterministic response times should remain locally enforced by turbine controllers or safety systems. The proposed protected session is intended for policy-mediated remote control or maintenance communication, where identity verification, authorization, and session revocation can be applied without placing the emergency shutdown path behind a new cryptographic session-establishment delay. Therefore, the session establishment delay should be bounded and validated against site-specific timing requirements before deployment.

The proposed approach also clarifies how Zero Trust communication can be adapted to OT constraints. A uniform strong-protection approach may improve security but can introduce unnecessary overhead in repetitive normal-state reporting. Conversely, a uniformly lightweight approach may reduce overhead but can fail to preserve important data during abnormal-state intervals. The proposed scheme avoids both extremes by using the operational state as the basis for selecting the required protection level. In this sense, the proposed approach does not simply add stronger protection to all communication, but applies protection selectively according to communication criticality. This is consistent with the Zero Trust principle that communication should not be implicitly trusted and that access or protection levels should be continuously determined according to context and policy.

In this prototype, the 4.1 MB test volume represents UDP-based telemetry reporting and scheme-specific retransmission or protection behavior, rather than a full encapsulation of industrial protocols such as MQTT, OPC UA, or Modbus. Nevertheless, the proposed state-aware filtering and protection logic is not tied to a specific protocol and could be implemented at offshore gateways or onshore monitoring platforms as a policy layer above or alongside such industrial communication protocols. From an enterprise information systems perspective, this allows state-aware OT communication to support onshore monitoring, maintenance management, asset management, and decision-support systems across multiple geographically distributed wind farm clusters.

Nevertheless, the numerical results should be interpreted as evidence of feasibility under controlled prototype-level assumptions, not as statistically generalizable results for all offshore wind farm OT

deployments. First, this case study does not reflect the full multivariate operational context of offshore wind farm OT, but is based on a single representative variable, generator bearing temperature. Second, the normal, warning, and critical thresholds used in this study are illustrative thresholds for explaining state-aware communication operation and do not represent universal safety standards or absolute industrial criteria. Third, the communication comparison was conducted under a single 5% packet-loss condition and the same topology. In addition, this prototype-level comparison does not include repeated trials or confidence intervals; therefore, the numerical results should be interpreted as an initial controlled comparison among protection strategies rather than statistically conclusive evidence.

Therefore, there are limitations in generalizing the results to diverse offshore communication environments, different network conditions, and different turbine configurations. However, the proposed three-state logic is not inherently limited to generator bearing temperature or UDP-based reporting, and it can be extended to multivariate state evaluation using variables such as vibration, rotor speed, and power output, as well as to other offshore communication patterns such as wireless links, cellular backhaul, or gateway-based offshore-to-onshore communication.

Accordingly, the contribution of this case study should be understood as an initial prototype-level demonstration that state-aware communication protection can be logically constructed and experimentally applied in offshore wind farm OT environments. The results show that differentiated protection can reduce unnecessary communication burden while preserving important warning-state communication and supporting restricted protected communication in the critical state. Future work should extend this validation to multivariate state assessment, multiple packet-loss conditions, diverse topologies, repeated trials, and more detailed measurements of critical-state session latency and protocol behavior.

5 Conclusion

This study began from the recognition that communications occurring in offshore wind farm OT environments do not all have the same meaning or require the same level of protection, yet prior discussions have not sufficiently distinguished among them. In particular, it pointed out the limitation that applying the same lightweight protection to all communications may lead to insufficient protection for important communications, whereas applying the same strong protection to all communications may increase unnecessary communication burden in a repetitive reporting structure. To address this issue, this study proposed a state-aware communication approach that classifies communication into the normal, warning, and critical states, interprets the meaning and importance of communication differently according to the operational state, and then applies differentiated protection levels accordingly.

The case study results showed that the proposed scheme maintained the loss rate of important communication in the warning state at the same level as r2, namely 0.3%, while the total communication volume was 4.1 MB, which was lower than r2's 6.1 MB. This indicates that the proposed scheme can maintain low-burden operation in the normal state while selectively reinforcing protection only in the warning state, thereby forming a more balanced protection structure between communication burden and the visibility of important communication.

However, this study has limitations in that it is a prototype-level case based on a single representative variable, the state classification criteria are also based on illustrative thresholds, and the experiment was conducted under a single packet-loss condition. Therefore, future research needs to extend the current single-variable threshold rule to a multivariate state-evaluation model that combines generator bearing temperature, rotor speed, vibration frequency, power output, and other condition-monitoring variables. Such a model could use statistical correlation or machine-learning-based classification to reduce false positives and blind spots caused by single-variable state transitions.

Additional validation under diverse communication environments, packet-loss conditions, and topology conditions is also required.

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